Simon Heilig

ELLIS PhD Student @ RUB & Cambridge
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Education

Since Oct. 2024	**Ph.D. Machine Learning** Ruhr-University Bochum, GER Area: Deep Learning for Spatiotemporal Data, Theory of Graph Neural ODEs Supervision: Prof. Fischer, Prof. Lio (Univ. of Cambridge, UK) ELLIS PhD Fellow, 2024
Apr. 2022 - May 2024	**M. Sc. Data Science** Friedrich-Alexander University Erlangen-Nürnberg, GER Thesis: "Hamiltonian Architectural Bias in Graph Neural Networks" Written while being visiting researcher with Prof. Bacciu (Univ. of Pisa, IT) Final Grade: 1.3 (3.7 GPA)
Oct. 2017 - Mar. 2021	**B. Eng. Computer Science** UAS Würzburg-Schweinfurt, GER Thesis: "Analysis and Revision of the MEKA Matrix Approximation Approach" Final Grade: 1.1 (4.0 GPA)

Professional Experience

Apr. 2023 - Oct. 2023	**Student Research Assistant** Friedrich-Alexander University Erlangen-Nürnberg Image Data Exploration and Analysis Lab (Prof. Kainz)
Oct. 2022 - Mar. 2023	**Student Teaching Assistant** Friedrich-Alexander University Erlangen-Nürnberg Stochastic Modeling (B. Sc. Mathematics); Chair for Stochastics (Prof. Krüger)
May 2021 - Nov. 2021	**Full-time Research Assistant** UAS Würzburg-Schweinfurt Supported by: ESF (WiT-HuB 4/2014-2020), Project KI-trifft-KMU
Dec. 2019 - Sep. 2022	**Student Teaching and Research Assistant** UAS Würzburg-Schweinfurt Computational Intelligence Group (Prof. Schleif); Teaching Assistant for Applied Numerics
Dec. 2018 - Feb. 2020	**Junior Software Engineer** Plunet GmbH Part-time Working Student and Full-time Intern (six months) as Java Backend Developer

Awards and Scholarships

2023	DAAD-PROMOS Travel Scholarship for Probabilistic AI Summer School
2022	Hans-Wilhelm Renkhoff Award for Outstanding Bachelor Thesis
2019	Max Weber-Program Scholarship of the Federal State of Bavaria
2018	Scholarship of the Federal Government of Germany

Selected Publications

Caralt, F., **Heilig**, **Simon**, A. Bazaga, A. Fischer, M. Eliasof, and P. Lio (2026). "Get a GRIP, this will be a long TRIP: A Quantifiable Long-Range Framework for Verifying Over-squashing". In: *The Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS)*. URL: https://openreview.net/forum?id=qTEOyobP0r.

Heilig, **Simon**, J. Püttschneider, M. Itani, A. Fischer, and T. Faulwasser (2026). "Port-Hamiltonian Neural Networks for Systems with Multiple Asymptotically Stable Equilibria". In: *NeurIPS Workshop: Axiom Foundations of Efficient Deep Learning*. URL: https://openreview.net/forum?id=cpziDxY4Ix.

Püttschneider, J., **Heilig**, **Simon**, A. Fischer, and T. Faulwasser (2026). "Towards the Optimal Control Foundation of Early Exits – Turnpikes and ResNets". In: *The Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS)*. URL: https://openreview.net/forum?id=anDvnfg9Dj.

Heilig, **Simon**, A. Gravina, A. Trenta, C. Gallicchio, and D. Bacciu (2025). "Port-Hamiltonian Architectural Bias for Long-Range Propagation in Deep Graph Networks". In: *The Thirteenth International Conference on Learning Representations (ICLR)*.

Münch, M., M. Röder, **Heilig**, **Simon**, C. Raab, and F. Schleif (2023). "Static and adaptive subspace information fusion for indefinite heterogeneous proximity data". In: *Neurocomputing* 555, p. 126635.

Heilig, **Simon**, M. Münch, and F. Schleif (2022). "Memory Efficient Kernel Approximation for Non-Stationary and Indefinite Kernels". In: *International Joint Conference on Neural Networks, IJCNN, Padova, Italy.*

Münch, M., **Heilig**, **Simon**, and F. Schleif (2021). "Multi-perspective embedding for non-metric time series classification". In: *29th European Symposium on Artificial Neural Networks, ESANN 2021, Bruges, Belgium.*